

Allen Xu

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EDUCATION

University of California, San Diego
Candidate for M.S. in Computer Science

Sep. 2026 - Present
San Diego, CA

Northeastern University - Khoury College of Computer Sciences
B.S. in Computer Science and Media Arts with Minor in Mathematics - Summa Cum Laude

Sep. 2022 - April 2026
Boston, MA

- GPA: 3.97/4.00
- Dean's Scholarship: Merit scholarship awarded in recognition of outstanding academic achievement

SKILLS

Languages: Java, Python, C++, TypeScript, Kotlin, SQL

Frameworks & Tools: React, Pytorch, Pandas, Git, REST API, OpenGL, Postman

Cloud & Data: AWS, PostgreSQL, Docker, Kubernetes

PROFESSIONAL EXPERIENCES

Resmed | *Java, DataDog, Kubernetes, Postman*
Software Engineer

May 2026 - Aug. 2026
San Diego, CA

- Enabled Japan-market integrators to consume ventilator data by designing a context-object pattern that propagated additional fields through both the legacy and modernized APIs, supporting 20M+ production API calls with no service disruption.
- Expanded Respiratory Care device support by upgrading the settings client schema and mapping 37 new fields, validating data accuracy against clinical metrics through Prism-proxy-based E2E tests.
- Led a Kubernetes namespace migration by identifying the optimal cutover window, reducing the message backlog 94% (93K → 6K)
- Improved cross-team testing and observability by building DataDog monitors and acceptance-test mocks adopted across 7 repositories

Insulet | *Python, Docker, PostgreSQL*
Devops Engineer

Jan. 2025 - May 2025
San Diego, CA

- Migrated live service Grafana database from SQLite3 to PostgreSQL, enabling scalability of **66+ tables** and an expanding user base.
- Containerized Grafana database services with Docker, improving modularity and deployment efficiency.
- Automated permission management by syncing LDAP's **2000+ employee** directory to Atlassian developer resources, reducing manual overhead and removing over **200+ redundant permission** grants currently added

MORSE Corp. | *Python, Docker, Airflow*
Software Engineer - Algorithmic Test and Evaluation

Jan. 2025 - May 2025
Boston, MA

- Implemented lazy loading into the test and analysis pipeline, enabling support for **2-3 million rows** of data and improving transformation speed; focused on pipeline modularity to account for shifting analyses up to higher security levels for further analysis.
- Optimized **20+ step** analysis pipeline by utilizing aggregated data representations, reducing server overhead and improving efficiency.
- Deployed Dockerized instances on virtual machines to access server side data, optimizing resource allocation and execution

Mirakl | *AWS, Python, Pandas, API*
Software Engineer - Data, AI and Analytics

Jan. 2024 - June 2024
Boston, MA

- Developed an XML sitemap parser to efficiently process **100K+ product entries**, replacing the company's reliance on Octoparse.
- Deployed cloud-based AWS applications continuously monitoring **50K+ product accounts**, enabling long term pattern detection.
- Worked with Mirakl's SQL product database to refine and augment **100k+ retailer product overlap** data including stock, prices, and product urls, served as a key factor in **30+ presentations** curated by the Sales team directed towards retailer CEOs and stakeholders
- Built a pipeline to match **10K Walmart sellers** to Amazon, used OpenAI's o4 model and MistralAI's Large model for data parsing and relevance evaluation to aid in a **\$1-2 million investment** decision by the Mirakl Labs team

Northeastern Khoury College of Computer Sciences | *Kotlin, Latex, Algorithms*
Teaching Assistant for Algorithms and Data Structures

Sep. 2023 - Dec. 2024
Boston, MA

- Provided personalized academic support through regularly-scheduled online one-on-one office hours to over **300 students**
- Gave feedback to professors in staff meetings on concepts that students find challenging based on overall student performance
- Planned and conducted weekly small-group lab sessions focused on reviewing challenging course concepts for groups of **30 students**

ACTIVITIES

NUAero | *C++, Git*
Redshift Propulsion Software Team Member

Sep. 2022 - June 2024
Boston, MA

- Developed data analysis functions to display critical data from sensors on the liquid engine onto the user interface in real-time
- Displayed fault readings from sensors to show and account for misalignment in the calibration of sensors and its measurements
- Managed and migrated Gitlab CI/CD pipeline for the engine control software to Github actions using Git Bash for **10 person team**